

Exploratory Data Analysis: Class Exercise

Task 1: Import Libraries and Load the Dataset

1. Import the necessary libraries (pandas, numpy, matplotlib, seaborn, scipy).
2. Load the Iris dataset.

Task 2: Basic Dataset Information

3. Display the first 5 rows of the dataset.
4. Use `.shape`, `.size`, `.columns` to display basic information

Task 3: Missing Values Check

5. Check if there are any missing values in the dataset. If any, identify how you would handle them.

Task 4: Descriptive Statistics

6. Use `.describe()` to calculate and display basic statistics (mean, standard deviation, min, 25th percentile, median, max) for all numerical features in the dataset.

Task 5: Grouping by Species

7. Group the dataset by the species column and calculate the mean for each feature within each species.

Task 6: Aggregation of Statistics

8. Use `.agg()` to calculate the mean, median, and standard deviation for each feature, grouped by species.

Task 7: Conditional Selection (Feature Filtering)

9. Select all rows where the sepal length is greater than the average sepal length of the dataset.

Task 8: Species with Specific Criteria

10. Filter and display the rows where the sepal width is greater than 3.0.

Task 9: Correlation Matrix

11. Calculate the correlation matrix between all numerical features and display it using a heatmap.

Task 10: Conditional Selection for Outliers

12. Identify and display rows where petal length is greater than 2 standard deviations from the mean.

Task 11: Visualize Distribution of Sepal Length

13. Plot the distribution of sepal length using a histogram.

Task 12: Boxplot of Petal Length by Species

14. Create a boxplot to compare the distribution of petal length across the three species.

Task 13: Probability Calculation of Species

15. Calculate the probability of an Iris flower being of a particular species based on the dataset.

Task 14: Conditional Probability Calculation

16. Calculate the conditional probability of an Iris flower being of species "Setosa" given that the sepal length is greater than 5.0.

Task 15: Histogram of Petal Length

17. Plot a histogram of the petal length feature with appropriate bin sizes.

Task 16: Scatter Plot of Sepal Length vs Petal Length

18. Create a scatter plot to visualize the relationship between sepal length and petal length.

Task 17: Grouped Statistics with Aggregation Functions

19. Use `.groupby('species').agg({'sepal length': 'mean', 'petal length': 'median'})` to calculate the mean of sepal length and the median of petal length for each species.

Task 18: Descriptive Statistics for a Specific Species

20. Filter the dataset for the species "Versicolor" and apply `.describe()` to summarize its statistics.

Task 19: Conditional Selection Based on Multiple Features

21. Select and display all rows where the sepal length is greater than 5.0 and petal length is less than 2.0.

Task 20: Skewness of Features

22. Calculate the skewness for each numerical feature in the dataset. Plot the skewness values for each feature using a bar plot.

Task 21: Variance of Features by Species

23. Calculate the variance of each feature for each species using `.groupby('species').var()`.

Task 22: Outlier Detection Using IQR

24. Identify and remove any outliers in the dataset using the Interquartile Range (IQR) method. Determine the lower and upper bounds for each feature and remove rows that fall outside these bounds.

Task 23: Outlier Detection Using Z-Scores

25. Calculate the Z-scores for all the numerical features in the dataset. Identify any outliers based on a Z-score threshold of 3 and remove them from the dataset.

Task 24: Handling Missing Data (if applicable)

26. If there are missing values in the dataset (detected in Task 3), handle them by either removing or imputing missing values, and explain the method used.